

SEM/EDS Polishing SOP — Vertical Cloud Lab

Standard operating procedure for sectioning, mounting, grinding, and polishing metallographic samples (primarily **aluminum / AISi10Mg**) to an SEM/EDS-ready finish.

Status: draft — working document. Transcribed from [byu-vcl #110](#) and not yet verified line-by-line against the equipment. Items marked **VERIFY** are known gaps — confirm on the machine and edit this file. This is **not** a substitute for the [BYU Engineering SOP template](#) paperwork or for in-person training on each machine.

- **Lab location:** polishing lab, [room # / building]
- **Primary contacts:** Gage Erickson, Ronnie Guymon
- **Equipment owner / trainer:** Project Support Center (PSC); Dr. Fullwood's group
- **Related:** [#110](#), [#77](#), [caliber #16](#) (vacuum chamber)

1. Safety and prerequisites

Emergency contacts (from the BYU SOP template):

- University Police: 801-422-2222
- Risk Management: 801-422-4468
- College Health & Safety Officer: 801-422-6589

Before you start

- Get trained in person on the saw, mounting press, and polisher before running any of them unsupervised.
- **Reserve the polisher.** There is a QR code on the machine that links to the reservation site. **VERIFY:** *record the reservation URL here.*
- Read the SDSs for ethanol, methanol, and acetone. Methanol is **toxic by skin absorption and ingestion**; all three are **flammable**.

PPE and controls

Hazard	Control
Flying debris, saw	Safety glasses at all times; keep the blade cover down
Solvents (ethanol, methanol, acetone)	Work in the fume hood ; nitrile gloves; no ignition sources
Hot mounting press	Do not touch the mold assembly until it has cooled
Sample contamination	Gloves whenever handling the puck after grinding starts

Hazard	Control
Rotating platen / belt	No loose sleeves, gloves clear of the belt, hair tied back

Waste: solvent rinses go to the labeled chemical waste container in the fume hood — never down the drain.

2. Consumables and materials

Item	Detail
Mounting compound	Bakelite (phenolic) — epoxy is the alternative for delicate samples
SiC grinding paper	320, 400, 600, 800, 1200 grit — VERIFY: the exact sequence stocked
Polishing pads	Adhesive-backed cloth, e.g. Imperial adhesive-back disc, 12 in / 300 mm
1 μm suspension	Allied High Tech Products, 1 μm alumina suspension, de-agglomerated
0.05 μm suspension	Allied High Tech Products #180-20000, 0.05 μm colloidal silica, non-crystallizing (vibratory polisher)
Solvents	Ethanol, methanol, acetone — in labeled squirt bottles
Parafilm	Bemis Parafilm M laboratory film — covers the solvent container in the ultrasonic cleaner
Kapton (polyimide) tape	For breaking the sample out of the Bakelite without damaging the surface
Silver conductive paint	SEM conductivity between sample and stub
Glassware	Deep glass containers (bowls / dishes) kept under the fume hood

Naming note: the 1 μm **alumina** and the 0.05 μm **colloidal silica** are different products used at different stages. Do not use the words interchangeably — see [§9 Open questions](#).

3. Sectioning — diamond saw

- Check out a blade.** Go to the Project Support Center and check out a diamond-edged saw blade. They give you a bag containing a couple of blades in a cardboard container, two wrenches, and some metal pieces (the flanges and nut). Take it to the saw in the polishing lab.
- Mount the blade.** Lift the platform to reach the arbor. Locate the bolt directly under and to the right of the slit in the platform where the blade passes through. Stack, in order, then

tighten with the two wrenches:

1. the wider flange ring, flat side toward the blade,
 2. the blade,
 3. the second wide flange ring, flat side toward the blade,
 4. the piece that fits into the recess of the flange ring,
 5. the nut.
3. **Water.** Confirm the water valve at the back is **closed** (handle parallel to the back of the machine). Fill the bucket from the fume hood and fill the machine until about $\frac{1}{4}$ - $\frac{1}{2}$ in of the blade sits in the water. Lower the platform back down.
 4. **Clamp the sample** against the two stabilizing rods, then clamp it in place. **VERIFY:** *describe / photograph the stabilizing rods.*
 5. **Check before starting:** blade cover down, saw not touching the sample.
 6. **Start the saw.** Switch on the right side toward the back → green button on the side → green button on the front. Bring the speed to about **1400 rpm**.
 7. **Cut.** Slowly rotate the handle to feed the blade into the sample, keeping the speed around **1000-1100 rpm** under load. Listen for unusual noise. A little water should be spraying up; **if no water is coming up, stop** — a dry blade will overheat and be ruined. Reposition as needed.
 - **Target size:** roughly **1 cm³** — small enough to sit on an SEM stub without overhanging the edges.
 8. **Clean up.** Disassemble the saw, open the water valve to drain, dry the machine and all saw hardware, and return the bag to the PSC.

4. Mounting — Bakelite press

The machine is a hot **compression mounting press** (it heats *and* presses). It is referred to in the lab as the "Bakelite press", "Bakelite oven", and "Bakelite compressor" — all the same machine.

1. Turn the press on with the switch at the back and open the water valve to the left of the machine (cooling water).
2. Press **Operate**, then set the parameters from the laminated sheet hanging above the machine. **VERIFY:** *copy those settings into this SOP so they survive the sheet going missing.*
3. Place the sample in the mold with the **face you want to polish pointing down**.
4. Lower the ram with the arrows on the screen.
5. Add $\frac{3}{4}$ of a **scoop** of Bakelite powder. **VERIFY:** *which scoop — the one stored with the machine?*
6. Fit the cap piece on top, **metal side down, green marble side up**, flush with the top of the machine. It has to be an airtight fit, so expect to wiggle it in.

7. Swing the lock bar over the chamber and press **Play**.
8. When the cycle finishes, remove the lock bar, press the **up arrow** to eject the puck, and pull it off the cap. It should release easily.
9. Turn the press off.

5. Rough grinding — belt sander

1. Turn on the belt sander and open the water valve on top so water runs down the belt.
2. Sand the puck by moving it **side to side** across the belt until the sample face is fully exposed and all Bakelite has been removed from the surface of interest.
3. Turn the sander off, then close **both** valves: the one on top of the sander and the large yellow-handled valve to the left of the machine.

VERIFY: belt grit, and how much pressure to use.

6. Grinding and polishing — automatic polisher

6.1 Machine check (before mounting the sample)

1. Turn the machine on and open its water valve.
2. Set the gentle values — **35, 30, 150** reading from the top of the screen down. **VERIFY:** units. Presumably force (N/lbf), head speed (rpm), and platen speed (rpm) — label them.
3. Set the timer to **30 seconds**.
4. Press **Link + H₂O** together to turn the water on, then **Link + Play** together to start it spinning.
5. If nothing is wrong, press **Stop**.

The **Link** button is in the top-right of the panel and acts as a modifier — hold it together with another key. **VERIFY:** the machine make/model and the real name of this button.

6.2 SiC grinding sequence

Cleanliness rule: rinse the puck, the metal puck holder, the sandpaper retaining ring, and the platen thoroughly **between every grit change**. A single carried-over coarse particle puts a scratch through everything you have done so far. Wear gloves and do not touch the sample face.

1. Press **Link + ↑** together to raise the puck holder.
2. Remove the splash-guard ring and the metal paper-retaining ring, lay down **320 grit** paper, and replace both rings.
3. Seat the puck in the metal holder with the sample face on the paper.

4. Turn on the water and start the platen as in §6.1. Position the water stream so it washes the removed particles **away from** the sample — if it washes them across the sample they will scratch it.
5. Grind while checking the surface frequently. **Move to the next grit only when every scratch on the face is the same size** (i.e. all the scratches from the previous grit are gone).
6. Rinse everything (see the cleanliness rule) and repeat for each grit:

Step	Grit	Notes
1	320	Until the face is flat and uniform
2	400	
3	600	
4	800	
5	1200	See the 1200-plain vs 1200-fine question in §9

VERIFY: *this sequence — the grits stocked may differ.*

All SiC papers are single use. If a paper starts to tear mid-run, stop and replace it.

6.3 1 μm alumina polish

1. **Pad.** Polishing pads may be reused **only on the exact same alloy** — reusing a pad across alloys cross-contaminates the surface.
 1. Peel the backing off to expose the adhesive side.
 2. Cut a small square of electroplating tape and stick it to the back of the pad, wrapping it around the edge to the top, to make a pull tab for removing the pad later. **VERIFY:** *the tape type — electroplating tape, electrical tape, or the Kapton tape we bought?*
 3. Stick the pad down on the platen.
2. **Prepare the puck and the landing zone before the machine is running:**
 - Blow the puck completely dry with compressed air from the fume hood and wipe the water off the **sides and bottom** with a paper towel.
 - **DO NOT WIPE THE SAMPLE FACE WITH THE PAPER TOWEL.**
 - Set out a clean paper towel to receive the puck after polishing, a deep glass container, and a squirt bottle of deionized water.
 - From this point on the puck may touch **only cleaned surfaces**, and the polished face is **never** set down.
3. **Charge the pad.** Apply **6-7 generous circles of 1 μm alumina suspension**, concentrated toward the center — the suspension migrates outward as the platen spins.
4. Swing the puck holder over the pad and press **Link + ↓** to lower it.
5. Set the timer to **3 minutes** and press start. **Do not run the machine's water during this step** — the alumina suspension is the lubricant and the abrasive; water would wash it

away.

6. **Keep the pad wet with alumina** throughout; you will see it moving toward the edge of the pad. Re-apply as needed.
7. **Between runs**, rinse the face with deionized water over the deep glass container and inspect for scratches. Repeat 3-minute runs until no scratches are visible. **VERIFY:** *how many runs is typical, and at what magnification are we judging "no visible scratches"?*
8. **Clean up.** Wash the alumina off the pad and off the whole machine in the fume hood. Blow the pad dry with compressed air, put the adhesive cover back on its sticky side, and bag it **labeled with both the abrasive (e.g. "1 μm alumina") and the alloy it was used on.**

7. Cleaning after polishing

The polished face must not sit in air while it is wet with suspension — silica starts to crystallize on the surface immediately.

7.1 After the alumina polish

1. Carry the puck to the fume hood **polished side up, in the deep glass container**.
2. Spray the face with ethanol and blow dry with compressed air.
3. **If a haze remains**, continue:
 1. Rinse the glass container twice with ethanol, emptying it into the chemical waste. The container and the puck are now both "clean".
 2. Place the puck face-up in the glass container and **submerge it completely in methanol** so the polished face is never exposed to air.
 3. Rinse the ultrasonic cleaner with deionized water, then fill it with $\frac{1}{4}$ – $\frac{1}{2}$ in of deionized water.
 4. Cover the glass container with a lid (or Parafilm) so no bath water can get in, and set it in the ultrasonic cleaner. Put the ultrasonic cleaner's own lid on and run it. **VERIFY:** *run time*.
 5. Take the container back to the fume hood. Remove the lid carefully so no bath water drips into the **methanol**.
 6. Blow dry with compressed air. Repeat the ultrasonication if the haze is still there.

7.2 After the vibratory polisher (colloidal silica)

This is the aggressive, recommended process. **Speed matters** — the sample must not sit in air between coming out of the colloidal silica and hitting the ethanol.

1. Prepare **3 glass bowls** (under the fume hood): rinse with water, dry with a paper towel.
2. Fill one with **acetone**, one with **ethanol**, one with **methanol** — just enough to cover the whole puck. Label or mark each bowl.

3. Take the ethanol squirt bottle and the ethanol bowl to the vibratory polisher and get ready to lift the puck and Vibromet holder out of the colloidal silica.
 4. Lift the sample out of the silica and **immediately start spraying it with ethanol**. While still spraying, free the puck from the holder and drop it into the ethanol bowl, **specimen side up**. Go to the ultrasonic cleaner.
 5. Move the puck from the ethanol to the **acetone**, cover the bowl with Parafilm, and ultrasonicate **10 minutes**.
 6. Move it back to the **ethanol**, Parafilm, ultrasonicate **8 minutes**.
 7. Move it to the **methanol**, Parafilm, ultrasonicate **5 minutes**.
 8. Remove from the methanol and dry **vigorously** with compressed air.
 9. Inspect the finish and repeat steps as needed.
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8. Vibratory polishing (colloidal silica)

Not yet written. The 0.05 μm colloidal silica step on the Vibromet comes between §6.3 and §7.2. Cleaning afterwards is §7.2, which is already documented.

9. Open questions and confusing parts

Points where this procedure is ambiguous, self-inconsistent, or incomplete. Resolve on the machine and edit this file.

1. **Alumina vs. silica in §6.3.** The original text says to use "1 μm alumina suspension" and then, one sentence later, to "apply 6–7 generous circles of silica". Those are two different products — the 1 μm alumina is the pad polish; the 0.05 μm colloidal silica is the vibratory-polisher medium. Written here as **alumina** throughout §6.3, on the assumption "silica" was a slip.
2. **§6.3 step order.** The original told you to lower the head onto the pad and *then* dry the puck and set out the clean towel / glass container / DI bottle. Reordered here so all preparation happens before the puck goes down.
3. **"3 minutes" vs. "every couple of minutes".** The original says to set 3 minutes, and also to rinse and inspect "every couple of minutes". Written here as repeated 3-minute runs with an inspection between each. Confirm.
4. **"DO NOT USE WATER" vs. the DI rinse.** These refer to different water: the machine's water jet stays off during polishing; the DI squirt bottle is used off the machine between runs. Stated explicitly here.
5. **Machine settings "35, 30, 150" have no units or labels.** Almost certainly force, head speed, and platen speed, but that needs confirming — and the machine make/model should be recorded so the next person can find the manual.

6. **The "Link" button.** Its real name and location should be confirmed; every control in §6 is a two-key combination with it.
7. **1200 plain vs. 1200 fine — is the fine one needed?** Check the labels first. If one reads **P1200** (FEPA grading) and the other **1200** with no P (CAMI/ANSI grading), they are *not* the same paper with a different texture — **P1200 $\approx$ 15 μ m** abrasive while **CAMI 1200 $\approx$ 3 μ m**, a 5 $\times$ difference. Practical answer: jumping from $\sim$ 15 μ m straight to the 1 μ m alumina works but costs a lot of polishing time; the $\sim$ 3 μ m paper bridges that gap cheaply. If both are the same grading standard and genuinely differ only in backing/texture, the plain one is sufficient — which matches what the PSC staff told us. **Action: read the two labels and record what they actually say.**
8. **Rotating the sample between grits.** Standard metallographic practice is to turn the sample 90° at each grit change so the new scratch direction is perpendicular — that is what makes "all scratches the same size" easy to judge. The original SOP does not mention it. Confirm whether the automatic head makes this moot (it rotates the sample continuously) or whether we should add it.
9. **Tape type in §6.3.** "Electroplating tape" may be a slip for electrical tape, or for the Kapton tape bought for this workflow ([order 12789](#)).
10. **§7.1 said "no water drips into ethanol"** at the point where the container holds **methanol**. Corrected here to methanol; confirm which solvent that final ultrasonication actually uses.
11. **Ultrasonication time in §7.1 is unspecified.** §7.2 uses 10 / 8 / 5 minutes for acetone / ethanol / methanol — pick a number for §7.1 and write it down.
12. **Kapton tape is bought but not in the procedure.** The plan was to use it to break the sample out of the Bakelite while preserving the surface. Add that step once the technique is settled.
13. **Bakelite press parameters live only on a laminated sheet** above the machine. Copy them into §4.
14. **Machine naming.** "Bakelite compressor", "Bakelite oven", and "the machine" are the same compression mounting press. Standardized here.
15. **Typo in the original:** "Return bad to PSC" $\rightarrow$ "return the **bag** to the PSC".
16. **Metal pucks instead of Bakelite.** Noted in #110 as a likely improvement — superglue the sample to a metal puck. If we switch, §4 is replaced entirely.
17. **Missing from this SOP:** PPE was not in the original (added in §1), and there is no mounting / stub-prep section covering the silver conductive paint and getting the sample into the SEM.

10. Revision history

Date	Change	By
2026-09-18	First draft, transcribed from #110	@ronnie-guymon (text), formatted by Claude